Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
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Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-07-14	Initial issue
2.0	2026-09-30	Reissued under INV-001: approval and execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
ADCC	antibody dependent cellular cytotoxicity
AMD	analytical method decision
AU	absorbance unit
BLA	Biologics License Application
CDC	complement dependent cytotoxicity
CEX	cation exchange chromatography
CPP	critical process parameter
CQA	critical quality attribute
CV	column volume
DNA	deoxyribonucleic acid
DoE	design of experiments
DS	drug substance
DV	diavolume
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G subclass 1
Init. RPN	initial risk priority number
KPP	key process parameter
LMW	low molecular weight species
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
NTU	nephelometric turbidity unit
OD280	optical density at 280 nm
PA	Protein A
pCO2	carbon dioxide partial pressure

Abbreviation	Meaning
QA	Quality Assurance
rcf	relative centrifugal force
RPN	risk priority number
SEC	size exclusion chromatography
Sev.	severity
SOP	standard operating procedure
UDP	uridine diphosphate
UF/DF	ultrafiltration and diafiltration
VCD	viable cell density
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

1 Purpose and scope

This document records the risk assessment made on the A-Mab drug substance process before any characterization study began. Its output is the scope of the characterization campaign. For each process parameter it names the quality attributes the parameter could affect, the potential failure mode and its effect, an initial risk score and the type of study assigned to the parameter. The assessment is prospective and reports no study result.

The assessment does not classify parameters. A parameter is classified as a critical process parameter (CPP), a well-controlled critical process parameter (WC-CPP), a key process parameter (KPP) or a general process parameter (GPP) from the effects that the characterization studies measure. That classification follows the parameter classification procedure SOP-4001. It is reported in the characterization report of each unit operation and consolidated in the master report PCMR-001. The scores in this document are initial scores in the same sense. The risk that remains once the studies have reported is a separate assessment, and nothing in this document anticipates it.

The scope is the drug substance process from the production bioreactor to formulation. Table 1.1 lists the 8 unit operations and the principal role of each, and the assessment covers their 37 process parameters. The seed expansion train, drug product manufacture and the analytical methods themselves are outside its scope.

Table 1.1: The A-Mab drug substance process train and the principal role of each unit operation.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate attributes
4	Harvest and Clarification	Primary recovery and clarification, where fragments form during the harvest hold
5	Protein A Chromatography	Capture; principal HCP and DNA clearance; source of leached Protein A
6	Low-pH Viral Inactivation	Low-pH hold; inactivates enveloped virus (XMuLV credit)
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; clears HCP, DNA and virus (XMuLV and MVM credit)
9	Small-Virus Retentive Filtration	Dedicated small-virus removal by size; principal MVM clearance (MVM and XMuLV credit)
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The Process Transfer Plan PTP-001 is the parent of this assessment. It moves the process from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI, and the parameters assessed here are those of the process description it transfers. The consumers of the assessment are the characterization plans PCP-003 to PCP-010, one for each unit operation. Each plan takes the study type assigned here and sets out how the study is run. Table 1.2 lists the documents to which this assessment is linked.

Table 1.2: Documents related to this assessment.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Applies this assessment at campaign level
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Master report of the characterization
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

In this framework risk assessment is a lifecycle activity, not a single exercise (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). This assessment is the first in that lifecycle. It draws on prior knowledge alone, because no A-Mab characterization data existed when it was made. It is revisited when the characterization reports are available, and again once commercial experience has been gained. The approach follows the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012) and supports Stage 1 of the process validation lifecycle (U.S. Food and Drug Administration 2011).

1.1 Changes in this revision

Version 1.0 of this assessment was approved on 2025-07-14. That date falls after the approval of PTP-001 on 2025-06-16 and before the approval of the Process Characterization Master Plan PCMP-001 on 2025-08-04. The first characterization study, S-BRX-SDM, the qualification runs of the production bioreactor scale-down model, started on 2025-08-11. Every study of the campaign started after this assessment was approved. When the Version 1.0 package was compiled, the title block of this assessment carried 2026-07-24, the date of the compilation and of the approval of the Version 1.0 reports, in place of its own approval date. The QA investigation INV-001 recorded this as part of its finding 5, which concerns the dates in the Version 1.0 records. QA opened the investigation on 2026-08-03, after its review of PCMR-001 Version 1.0 for readiness for the Biologics License Application (BLA), and closed it on 2026-09-18. Its corrective action CA3 reissues the characterization reports, PTP-001,

this assessment and PCMR-001 at Version 2.0. This assessment is reissued under it, effective 2026-09-30.

This revision makes the following changes.

1. It records the approval date of Version 1.0 in the title block and in the revision history, and this section records the date on which the first characterization study started.
2. It gives the quality attribute register as it now stands (Table 3.1). The register adds fragments, the low molecular weight species formed when the antibody is clipped. PCMP-001 Version 2.0 §7.8 sets non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS) as the method for fragments, following the decision of 2026-03-20 to change the fragment method (AMD-001). Section 3 gives the origin of their criterion and what the addition means for this assessment. The register also prints each criterion by its type, as a lower limit, an upper limit or a range. Version 1.0 printed both ends of every criterion, so that viral clearance read as if it had an upper limit.
3. Where this document refers to acceptance criteria or to scale-down models, it refers to PCMP-001 Version 2.0. That plan qualifies each scale-down model by its offset from manufacturing scale (PCMP-001 §7.1). It judges each step output on which a drug substance criterion depends against an intermediate acceptance criterion (PCMP-001 §7.3).
4. It corrects the descriptions of Version 1.0 that were wrong (Table 1.3 and Table 1.4).

Table 1.3: Descriptions of Version 1.0 corrected in Version 2.0, Sections 2 to 4.4. No score and no study type changes with them.

Section	Version 1.0	Version 2.0
2 and 5	A parameter that puts no quality attribute at risk is studied one at a time, and so are the lowest ranked parameters.	Such a parameter is studied one factor at a time unless it is carried into the design of its step, as 2 Protein A parameters are.
2 and 6	Severity separates the parameters of a unit operation, and the RPN does not separate parameters within a class.	Within a class of parameter the RPN differs only through severity. Between classes, occurrence and detection differ as well.
2	Criticality as a continuum cited to ICH Q9(R1).	Cited to ICH Q8(R2), as in PCMP-001 Version 2.0 §2.
3	Viral clearance takes its severity from the consequence of a shortfall and not from the probability of detecting one.	Its severity follows from the very high criticality that the register gives viral clearance by rule. That no release test detects a shortfall enters the detection score.
4 and 5	Each subsection gives the interaction judgement that decides the study type, recorded per parameter.	Section 4 records the judgement where it decides the study type.
4.1	The production bioreactor carries the highest severity of any step.	Culture pH is the only parameter outside the steps with a viral clearance claim that has a severity of 10.
	Acidic charge variants fall as culture pH rises.	Acidic variants are expected to rise with culture pH, because deamidation is catalysed by base.
	Temperature and pH act on the same glycosyltransferase activity.	Both act on the same glycan processing enzymes, the mannosidases and the galactosyltransferase.
	Dissolved oxygen raises titre through the productivity of the cells and not through their number.	Dissolved oxygen acts on peak viable cell density, titre and process consistency, as Table 4.2 gives it.
4.1 to 4.8, Tables 4.2 to 4.9	Failure-mode texts with the labels and units of Version 1.0 (Dissolved CO2, mOsm, titer), the header CQA(s) at risk, and the UF/DF buffer exchange called not part of drug substance characterization.	The parameter names and units of the rest of the package, the header Attribute(s) at risk, since Section 3 calls only high and very high attributes critical, and the UF/DF phrase removed.
4.3	The 2 Protein A parameters scored on step yield cost no additional runs in the design.	Withdrawn. They are carried into the design so that the same design bounds step yield.
4.4	Inactivation of enveloped virus at low pH follows first order kinetics.	Close to first order over most of the hold, often with tailing near the limit of detection.
	A hold longer than its range raises aggregate at any pH in the range.	An expectation from platform experience. No study had been run.

Table 1.4: Descriptions of Version 1.0 corrected in Version 2.0, Sections 4.5 to 6. No score and no study type changes with them.

Section	Version 1.0	Version 2.0
4.5	Protein load carries the highest severity of the step because aggregate and host cell protein are both at risk.	Protein load, elution buffer pH and elution stop collect share the severity of aggregate.
4.6	At the load pH, host cell protein, DNA and virus carry a net negative charge.	DNA, virus and most host cell proteins do. Basic host cell proteins pass through with the antibody.
4.6, Table 4.7	Protein load: reduced HCP removal capacity, with viral clearance robust.	The words on viral clearance are removed. They conflicted with the risk that the binding capacity of the resin is exceeded.
4.7	Prior knowledge places neither the size nor the direction of a pressure effect on log reduction.	Parvovirus filters can retain less virus at low pressure or low flux and after a pressure interruption. The direction is known, and its size for this filter is not.
4.7, Table 4.8	Volumetric load above 105 L/m2, the upper limit of the NOR.	Above 140 L/m2, the upper end of the range studied.
4.8	UF/DF forms and clears no quality attribute.	It forms and clears no attribute in Table 3.1. Its final concentration is released against a specification.
5	A minimum hold time and a maximum volumetric load share a bound with their NOR.	The low end of the range studied is shared with the NOR by 5 parameters, among them hold time and filtration volume.
	The assignment is the output of this assessment.	The main output. Section 7 lists the outputs.
6	A severity says what a failure would mean for the patient.	These words are dropped, and the rest of the sentence is kept.
	If the anion exchange study finds that residence time acts together with the load chemistry, the design would have to be augmented.	No study as assigned can detect that interaction. The univariate result bounds the step only if the interaction is absent, as prior knowledge expects.
	Each parameter is assessed against the attributes of its own step.	Each parameter is assessed against the attributes it can affect at its own step.

The method in Section 2, the scores and the ranking in Section 4 and the study assignment in Section 5 are unchanged. They are those approved on 2025-07-14, before the studies. No study result enters this assessment, and it gives no final classification of any parameter.

2 Risk assessment methodology

The method is risk ranking and filtering, applied to process parameters as in the A-Mab case study (CMC Biotech Working Group 2009) and within the quality risk management principles of ICH Q9 (International Council for Harmonisation 2023b). Each parameter is put to two questions. The first asks what the parameter could do to a quality attribute, or to process performance, if it moved outside the range over which the process is currently understood. The second asks whether the effect of the parameter could depend on the setting of another parameter of the same unit operation. The first question sets the score. The second sets the study type, because an effect that depends on a second parameter cannot be found by moving one parameter at a time.

Severity is taken from the attribute at risk and not from the parameter. It runs from 2 to 10 and is the severity assigned to the attribute in the quality attribute assessment, reproduced in Table 3.1. Where a parameter could affect more than one attribute, the severity of the most severe of them is used. A parameter that puts no quality attribute at risk takes a severity of 4, the value assigned to a process attribute such as step yield or titre.

Occurrence and detection are assigned before any effect has been measured, and they are assigned by class and not parameter by parameter. Table 2.1 gives each score with its basis. Occurrence is scored at 7 for a parameter that could affect a quality attribute, because neither the size of its effect nor the width of the range that holds the attribute in limit is known at this point. It is scored at 4 for a parameter that puts no quality attribute at risk, where the platform history bounds how often the parameter leaves its range. Detection is scored on what the existing controls would reveal. A failure that acts on viral clearance is scored at 10, since no in-process control and no release test on the drug substance measures the log reduction a step achieves. A failure that changes another quality attribute is scored at 8, since release testing measures the attribute but monitoring during the unit operation does not. A failure that puts no quality attribute at risk is scored at 6, since it shows up in the performance of a downstream unit operation before the batch is released.

Table 2.1: Scores used for the initial risk of each parameter, by class of parameter.

Score	Parameter class	Value	Basis
Severity	Quality attribute at risk	2 to 10	Severity of the most severe attribute at risk, from the quality attribute assessment
	Process performance only	4	The value assigned to a process attribute such as step yield or titre
Occurrence	Quality attribute at risk	7	Neither the size of the effect nor the width of the range that holds the attribute in limit is known
	Process performance only	4	The platform history bounds how often the parameter leaves its range
Detection	Viral clearance at risk	10	No in-process control and no release test on the drug substance measures the log reduction a step achieves
	Other quality attribute at risk	8	Release testing measures the attribute, but monitoring during the unit operation does not
	Process performance only	6	Shows up in the performance of a downstream unit operation before the batch is released

The initial risk priority number (RPN) is the product of severity, occurrence and detection. Across the assessment it runs from 96 to 700 and takes 5 distinct values, because occurrence and detection are assigned by class. Within a class of parameter the RPN therefore differs only through severity. Priority is High where an attribute of severity 8 or above is at risk, Medium where a quality attribute of lower severity is at risk, and Low where only process performance is at risk. No parameter is assigned a study type on its RPN alone.

The assessment uses 3 study types. A parameter carried into the designed experiment of its unit operation is studied multivariately, which is the only way to estimate an effect that depends on a second parameter. A parameter that could affect a quality attribute but is not carried into the design is studied one factor at a time, and the reason it was left out of the design is recorded with it. The A-Mab risk ranking and filtering approach permits this substitution where the study can be justified (CMC Biotech Working Group 2009), and the assessment marks such a parameter "univariate (justified)". A parameter that puts no quality attribute at risk is studied one factor at a time unless it is carried into the design of its step (Section 5). The A-Mab approach also allows a parameter to be given no new study where prior knowledge supports its range. This assessment assigns no parameter to that outcome. Each characterization plan states how the assigned study is executed.

Criticality is treated as a continuum and not as a binary split (International Council for Harmonisation 2009; CMC Biotech Working Group 2009). The ranking in Table 4.1 therefore orders parameters against each other. It does not divide them into a critical and a non-critical group, and it is not a control strategy. Both follow from the studies that this assessment scopes.

3 Quality attributes at risk

The severity axis of the assessment comes from the quality attribute register. Table 3.1 gives the 11 attributes of the drug substance as the register now stands. For each attribute it gives the acceptance criterion, the criticality, the severity carried into this assessment and the unit operation that forms the attribute. For the two cumulative viral clearance requirements it gives the unit operation to which the register assigns the requirement. The criteria are those of the A-Mab case study, except the criterion for fragments (CMC Biotech Working Group 2009). Criticality is the product of impact and uncertainty, which is Tool #1 of the A-Mab attribute assessment. Attributes of high and very high criticality are called critical, and 6 of the 11 attributes are critical on that definition. The two viral clearance attributes carry very high criticality by rule, although their Tool #1 score is 20, because a viral contamination would harm the patient directly (PCMP-001 §3).

Table 3.1: Drug substance quality attributes as the register now stands, with their criticality and the severity carried into this assessment.

Category	Quality attribute	Acceptance	Criticality	Tool #1 score	Severity	Formed or set at
glycosylation	Afucosylation	2–13 % of glycans	H	60	8	Production Bioreactor
	Galactosylation (total %Gal)	10–40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
	High Mannose	3–10 % of glycans	VH	80	10	Production Bioreactor
size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60	8	Production Bioreactor
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)[a)]	L	12	—[b)]	Harvest and Clarification
charge variant	Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36	6	Production Bioreactor
	Residual DNA	≤ 0.001 ng/dose[c)]	VL	6	2	Production Bioreactor
	Leached Protein A	≤ 5 ppm	L-M	16	4	Protein A Chromatography
viral safety	Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation[d)]
	Viral clearance — MVM (parvovirus)	≥ 8.6 log10 (cumulative)	VH	20	10	Anion Exchange Chromatography[d)]

[a)] Set by the sponsor when fragments joined the register at Version 2.0. The other criteria are those of the A-Mab case study.
[b)] Not scored. Fragments joined the register after this assessment was made, and no severity is assigned to them.
[c)] Per dose of 10 mg/kg, the dose of the A-Mab case study, for a reference adult of 70 kg, which is 700 mg of antibody. The body weight is an assumption.
[d)] The unit operation to which the register assigns the cumulative requirement.
Criticality levels are VL very low, L low, L-M low to moderate, M-H moderate to high, H high and VH very high.

Of the 11 attributes, 7 are formed in the production bioreactor. The register assigns fragments to harvest, leached Protein A enters the process at the capture step, and each viral clearance requirement is assigned to one of the steps that clear virus. The bioreactor carries more parameters into multivariate study than any other unit operation (Table 5.1).

The highest severity, 10, is carried by 3 attributes. High mannose has the highest Tool #1 score in the register, 80. High mannose glycans carry no core fucose, so they are expected to raise antibody dependent cellular cytotoxicity (ADCC) in the same way as afucosylation, and A-Mab acts on its target through ADCC. The impact of the attribute is therefore scored on efficacy. Its uncertainty is scored above that of afucosylation, because the link rests on published literature and not on assays run with A-Mab.

Both viral clearance attributes carry the highest severity for a different reason. Each

requirement is cumulative across the steps that clear the virus, which are low-pH viral inactivation, anion exchange and virus filtration for XMuLV, and anion exchange and virus filtration for MVM. The register assigns each requirement to one step, and the claims of these steps add up to it (PCMP-001 §7.4). A shortfall at one of these steps changes no property of the drug substance that a release test measures (International Council for Harmonisation 2023a). Their severity follows from the very high criticality that the register gives viral clearance by rule (PCMP-001 §3). That no release test detects a shortfall enters the detection score (Table 2.1), not the severity.

Acidic charge variants sit at the other end of the register, with a Tool #1 score of 4 and a severity of 4. Their impact is scored low because deamidation of A-Mab occurs at sites outside the binding region and outside the part of the Fc region that carries effector function, so the deamidated forms keep their activity. The assessment still scores 4 bioreactor parameters against acidic variants. The range of each parameter over which acidic variants stay within their criterion is one of the things the bioreactor study must establish. Residual DNA, with the lowest severity, 2, is treated the same way. A low severity lowers the priority of a parameter and does not remove it from study.

Fragments are new to the register at Version 2.0. The A-Mab case study lists fragmentation among the typical quality attributes of an antibody but gives it no limit, and the sponsor set the criterion for fragments, ≤ 2 % (nrCE-SDS), when they joined the register (PCMP-001 §3). Their criticality is low, with a Tool #1 score of 12. The register assigns fragments to harvest, because proteases released from lysed cells clip the antibody in culture and during the harvest hold.

This assessment was made before fragments joined the register. No parameter is scored against fragments and no severity is assigned to them. The study assignment in Section 5 was made without them. The harvest hold, where fragments form, is not a parameter of the list assessed here (Section 6). Whether a parameter could change fragments is therefore outside this assessment.

4 Parameter risk ranking by unit operation

Table 4.1 gives the ranking for the whole train. Of the 37 parameters, 21 are scored against at least one quality attribute and 16 against process performance alone. The ranking puts 16 parameters at High priority, 5 at Medium and 16 at Low. The highest RPN, 700, is reached by 8 parameters, all of them at the steps with a viral clearance claim and all of them scored against viral clearance.

The subsections that follow take each unit operation in turn. Each gives the potential failure mode and effect behind each score and, where it decides the study type, the judgement on interactions.

Table 4.1: Pre-characterization risk ranking of the process parameters, by unit operation. The scores are initial scores, and no residual risk is reported.

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Priority
3	Production Bioreactor	Culture pH	10	560	multivariate DoE	High
		Culture temperature	8	448	multivariate DoE	High
		CO2 partial pressure (pCO2)	8	448	multivariate DoE	High
		Osmolality	8	448	multivariate DoE	High
		Culture duration	8	448	multivariate DoE	High
		Dissolved oxygen	4	96	univariate	Low
		Initial viable cell conc.	4	96	univariate	Low
		Basal medium concentration	4	96	univariate	Low
		Nutrient feed-1 volume	4	96	univariate	Low
4	Harvest and Clarification	Centrifugation (rcf)	4	96	univariate	Low
		Depth filter load	4	96	univariate	Low
		Post-clarification turbidity	4	96	univariate	Low
5	Protein A Chromatography	Protein load	6	336	multivariate DoE	Medium
		Elution buffer pH	6	336	multivariate DoE	Medium
		Load flow rate	4	96	multivariate DoE	Low
		End of pool collect	4	96	multivariate DoE	Low
		Operating temperature	4	96	univariate	Low
		Bed height	4	96	univariate	Low
6	Low-pH Viral Inactivation	Inactivation pH	10	700	multivariate DoE	High
		Hold time	10	700	multivariate DoE	High
		Temperature	10	700	multivariate DoE	High
		A-Mab concentration	4	96	univariate	Low
7	Cation Exchange Chromatography	Protein load	8	448	multivariate DoE	High
		Load/Wash conductivity	6	336	multivariate DoE	Medium
		Elution buffer pH	8	448	multivariate DoE	High
		Elution stop collect	8	448	multivariate DoE	High
		Elution flow rate	4	96	univariate	Low
8	Anion Exchange Chromatography	Load pH	10	700	multivariate DoE	High
		Equil/Wash-1 conductivity	6	336	multivariate DoE	Medium
		Load conductivity	10	700	multivariate DoE	High
		Protein load	6	336	multivariate DoE	Medium
		Operating flow rate	10	700	univariate (justified)	High
9	Small-Virus Retentive Filtration	Filtration volume (load)	10	700	multivariate DoE	High
		Filtration pressure	10	700	multivariate DoE	High
10	Ultrafiltration / Diafiltration	Number of diavolumes	4	96	univariate	Low
		Transmembrane pressure	4	96	univariate	Low
		Final DS concentration	4	96	univariate	Low

Sev., severity of the most severe attribute at risk. Init. RPN, initial risk priority number, the product of severity, occurrence and detection.

4.1 Production Bioreactor

The production bioreactor forms the glycan, charge variant and aggregate attributes of the drug substance. Of its 9 parameters, 5 are scored against a quality attribute, all of them at a severity of 8 or above. Culture pH is the only parameter outside the

steps with a viral clearance claim that has a severity of 10, and it has the highest RPN in the step, 560.

Culture pH and culture duration are each scored against 5 attributes, more than any other parameter in the assessment. Culture pH takes a severity of 10 because high mannose is one of the attributes it puts at risk. A culture pH above its range is expected to raise high mannose and to lower galactosylation. A higher pH in the medium raises the pH in the Golgi, while the mannosidases and the galactosyltransferase work fastest under more acidic conditions. Less mannose is then trimmed from the glycans and less galactose is added to them. Aggregate in the harvest is expected to rise with culture pH and afucosylation to fall. Acidic variants are expected to rise with culture pH, because the deamidation of asparagine residues is catalysed by base.

Culture duration reaches as many attributes by a different route. A longer culture, or one harvested after viability has fallen, is expected to raise host cell protein and DNA in the harvest, because lysing cells release their protein and DNA into the medium. Aggregate is expected to rise with duration too, since antibody secreted early in the run is held at culture temperature until harvest. Galactosylation is expected to fall late in the culture as the supply of UDP-galactose to the galactosyltransferase runs down. Afucosylation is expected to fall with duration as well, and the study has to bound that fall against the lower limit of the afucosylation criterion.

Culture temperature, pCO2 and osmolality are each scored against galactosylation and acidic variants, and temperature is also scored against afucosylation. Galactosylation is expected to rise with culture temperature, which raises the activity of the galactosyltransferase, and to fall as pCO2 and osmolality rise. Temperature is expected to act together with pH on the glycan attributes, because both act on the activity of the same glycan processing enzymes, the mannosidases and the galactosyltransferase. An effect of that kind is estimated only by moving the two parameters together. The 5 parameters scored against a glycan, charge variant or aggregate attribute are therefore carried into one designed experiment.

The other 4 parameters change how many cells the culture reaches and how much antibody each cell makes, and none of them is scored against a quality attribute. Initial viable cell concentration, basal medium concentration and nutrient feed-1 volume each raise the integral of viable cell concentration over their ranges, and titre rises with it. Dissolved oxygen acts on peak viable cell density, titre and process consistency. Basal medium concentration needed an argument. Platform experience shows a statistically significant but shallow effect on afucosylation, too small to move the attribute across its criterion over the concentrations the process uses. The parameter is therefore scored on process performance and studied one factor at a time. Nutrient feed-1 volume is treated in the same way, since platform experience shows only a slight effect on galactosylation. Table 4.2 gives the failure modes of the step.

Table 4.2: Potential failure modes and their effects, production bioreactor.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Culture pH	Culture pH outside its characterized range (pH 6.6–7.1)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)
Culture temperature	Culture temperature outside 34–36 °C	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)
CO2 partial pressure (pCO2)	CO2 partial pressure (pCO2) outside 40–160 mmHg	Strong effect on galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Osmolality	Osmolality outside 360–440 mOsm/kg	Affects galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Culture duration	Culture duration beyond 19 days, or harvest at low viability	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA
Dissolved oxygen	Dissolved oxygen outside 30–70 % air saturation	Impacts peak VCD, titre and process consistency (no significant CQA impact)	process performance
Initial viable cell conc.	Initial viable cell conc. outside 0.5–1.5 × 10^6 cells/mL	Drives peak/integral VCD and titre; no product-quality impact	process performance
Basal medium concentration	Basal medium concentration outside 0.8–1.6 × standard	Statistically significant but shallow effect on afucosylation; drives titre	process performance
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9–15 % of working volume	Slight effect on galactosylation; critical for titre	process performance

4.2 Harvest and Clarification

Harvest separates the cells from the culture and clarifies the supernatant for capture. Its 3 parameters are scored against process performance alone, and all of them sit at the floor of the ranking at 96. No parameter of this step is carried into a designed

experiment. The A-Mab case study assumes, for simplicity, that harvest has no impact on product quality (CMC Biotech Working Group 2009). The register as it now stands assigns fragments to this step, which the assessment did not score (Section 3).

The failure modes are a centrifugal force outside its range, a depth filter loaded beyond its capacity and a high turbidity after clarification. A centrifugal force below its range leaves more cells and cell debris in the centrate. A particle settles faster the stronger the field acting on it, so a weaker field removes less of the fine debris in the time the material spends in the bowl. That debris then loads the depth filters. A depth filter loaded beyond its capacity passes particulate instead of retaining it, and the turbidity of the clarified harvest rises. Turbidity is measured on the clarified harvest, and particulate that reaches the capture column raises its back pressure. A failure at this step is therefore seen before the batch is released. Table 4.3 gives the failure modes.

Table 4.3: Potential failure modes and their effects, harvest.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Centrifugation (rcf)	Centrifugation (rcf) outside 6000–12000 g	Affects clarification/yield (no CQA impact)	process performance
Depth filter load	Depth filter load outside 80–160 L/m2	Affects filter capacity/turbidity (no CQA impact)	process performance
Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)	process performance

4.3 Protein A Chromatography

Capture binds the antibody from the clarified harvest and removes most of the host cell protein and DNA. Protein load and elution buffer pH are the 2 parameters scored against a quality attribute, both against host cell protein at a severity of 6. A protein load above its range is expected to raise host cell protein per milligram of antibody in the pool, because host cell protein that binds to the captured antibody is not removed by the wash, and more of it is expected to be bound per milligram of antibody on a more heavily loaded column. A lower elution pH is expected to raise it as well, since a more acidic elution buffer releases host cell protein that the wash left bound to the resin. The two parameters are expected to act together, and both are carried into the capture design.

The capture eluate is the load of the low-pH hold. Elution buffer pH therefore sets the pH of the pool before titration and the amount of acid needed to bring it to the hold set-point.

Load flow rate and end of pool collect are scored against step yield and against no quality attribute. Yield is expected to fall when a high load is run at a high flow rate, because the antibody has less time to diffuse into the resin pores and part of it passes

through the column unbound. Yield is expected to rise as collection is carried further down the tail of the elution peak, up to the point where the pool is diluted without gaining product. Both parameters are carried into the capture design, so that the same design bounds step yield. Operating temperature and bed height are studied one factor at a time.

No operating parameter of this step is scored against leached Protein A, although the attribute enters the process here. Leached Protein A comes from the affinity ligand itself. This assessment attributed the amount released to the resin and its history of use, which are controlled under the resin life cycle procedure SOP-2008, and not to a parameter in Table 5.2. The polishing steps clear what reaches them. The capture study measures leached Protein A in the pool across the designed experiment, so the ranges this assessment scopes are bounded against it even though no parameter is ranked on it. Table 4.4 gives the failure modes.

Table 4.4: Potential failure modes and their effects, Protein A.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Protein load	Protein load above its characterized range (above 50 g/L resin)	Increases pool HCP (interaction with elution pH)	Host Cell Protein (HCP)
Elution buffer pH	Elution buffer pH below its characterized range (below pH 3.2)	Increases pool HCP; also sets up viral inactivation feed	Host Cell Protein (HCP)
Load flow rate	Load flow rate high, with a high protein load	Reduces product yield (no CQA impact)	process performance
End of pool collect	End of pool collect outside 2–3.2 CV	Affects product yield (no CQA impact)	process performance
Operating temperature	Parameter outside characterized range	No significant product-quality impact expected	process performance
Bed height	Parameter outside characterized range	No significant product-quality impact expected	process performance

4.4 Low-pH Viral Inactivation

Holding the pool at low pH inactivates enveloped virus by disrupting the lipid envelope that XMuLV needs to enter a cell. The register assigns the cumulative XMuLV clearance requirement to this step. Of its 4 parameters, 3 reach 700, the highest RPN in the assessment.

Inactivation of enveloped virus at low pH is close to first order over most of the hold, and it often tails as the titre approaches the limit of detection. Inactivation pH and temperature set the rate constant, and the hold time sets how long that rate acts. Where the inactivation is first order, the log reduction is proportional to the rate

constant integrated over the hold. A pH above its range lowers the rate constant and leaves less virus inactivated in the same hold. A hold shorter than its range leaves less virus inactivated at the same pH. A temperature below its range lowers the rate constant too, and this matters most at the short end of the hold and the high end of the pH range, where the log reduction reached is smallest. The 3 parameters are studied in one designed experiment, because the log reduction depends on all of them together and none can be bounded with the others held at one setting.

Inactivation pH has a second failure mode in the opposite direction. A pH below its range exposes the antibody to conditions under which aggregation and precipitation have been seen on the platform. A hold longer than its range is expected, from platform experience, to raise aggregate at any pH in the range. Inactivation pH and hold time are therefore scored against aggregate as well.

Detection scores 10 for each of these parameters. Neither an in-process control nor a release test of the drug substance measures the log reduction that the hold achieves. The viral clearance claim of the step is made instead from a spiking study in a qualified small-scale model (International Council for Harmonisation 2023a).

A-Mab concentration is scored against process performance. Prior product experience showed no effect of concentration on the rate of inactivation, on aggregation or on charge variants over the range the pool reaches. The parameter is therefore studied one factor at a time. No parameter of this step is scored against MVM, because a parvovirus has no lipid envelope for low pH to disrupt (PCMP-001 §7.4). Table 4.5 gives the failure modes.

Table 4.5: Potential failure modes and their effects, low-pH viral inactivation.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Inactivation pH	Inactivation pH above pH 4 (incomplete inactivation) or below pH 3.2 (aggregation or precipitation)	Loss of XMuLV inactivation assurance or increased aggregate	Viral clearance — XMuLV (enveloped), Aggregates (HMW)
Hold time	Hold time below 60 min (incomplete inactivation) or above 180 min (aggregation)	Incomplete inactivation or increased aggregate	Aggregates (HMW), Viral clearance — XMuLV (enveloped)
Temperature	Temperature below 15 °C	Lower inactivation rate at short time / higher pH	Viral clearance — XMuLV (enveloped)
A-Mab concentration	A-Mab concentration above 35 g/L	No significant effect on inactivation, aggregation or charge variants	process performance

4.5 Cation Exchange Chromatography

The cation exchange step is the principal aggregate reduction step of the process. It also clears host cell protein, DNA and leached Protein A, and it forms no attribute of its own. Of its 5 parameters, 4 are scored against a quality attribute. Protein load, elution buffer pH and elution stop collect are scored against aggregate at a severity of 8, and protein load is scored against host cell protein as well.

Aggregate is the attribute that matters most here. No step after cation exchange is credited with aggregate reduction, so aggregate that leaves this step in the pool reaches the drug substance. Pool aggregate is expected to rise with protein load, elution buffer pH and elution stop collect. Aggregate elutes after the monomer, so collecting further down the tail of the elution peak takes more of it into the pool. A higher load leaves less resin per gram of antibody to separate the two. Load and elution pH are expected to act together, since the resolution between monomer and aggregate depends on both. The conductivity of the load and wash is scored against host cell protein. A conductivity below its range weakens the wash and leaves more host cell protein bound, to be eluted with the product.

Elution flow rate is scored against step yield. A higher elution flow rate broadens the elution peak, because the antibody has less time to diffuse out of the resin pores, and a broader peak dilutes the pool. Prior knowledge places no effect of flow rate on pool aggregate large enough to score, so the parameter is studied one factor at a time. Table 4.6 gives the failure modes.

Table 4.6: Potential failure modes and their effects, cation exchange.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Protein load	Protein load above its characterized range (above 40 g/L resin)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	Aggregates (HMW), Host Cell Protein (HCP)
Load/Wash conductivity	Load/Wash conductivity below 3 mS/cm	Reduced HCP clearance	Host Cell Protein (HCP)
Elution buffer pH	Elution buffer pH outside pH 5.8–6.2	Alters aggregate distribution and yield	Aggregates (HMW)
Elution stop collect	Elution stop collect above 1.5 AU (OD280, descending), a later stop on the descending peak	Minor increase in pool aggregate	Aggregates (HMW)
Elution flow rate	Elution flow rate outside 100–300 cm/hr	Affects peak shape and yield (minor aggregate)	process performance

4.6 Anion Exchange Chromatography

The anion exchange step is run in flow-through mode. At the load pH, DNA, virus and most host cell proteins carry a net negative charge and bind to the resin. The load pH lies below the isoelectric point of A-Mab, which therefore carries a net positive charge and passes through, together with the basic host cell proteins. The register assigns the cumulative MVM clearance requirement to this step (Section 3). All 5 of its parameters are scored against a quality attribute.

Load pH is scored against 3 attributes and reaches 700. It sets the charge on the impurities and on the virus, so a load pH below its range weakens the binding of host cell protein and of both model viruses at once. Clearance is also expected to fall as load conductivity rises, because the counter-ions in the load compete with virus and impurity for the charged groups of the resin. Load pH and load conductivity are expected to act together, since one sets the charge on each species and the other sets how strongly that charge is screened. The conductivity of the equilibration and wash-1 buffers and the protein load are scored against host cell protein. A higher wash conductivity weakens the binding of host cell protein in the same way. A higher protein load presents more impurity to a fixed volume of resin, so its binding capacity can be exceeded.

Operating flow rate is the one parameter in the assessment with a quality attribute at risk that is not carried into a designed experiment. A higher flow rate shortens the residence time of the load in the bed, which gives a virus particle less time to reach and bind a charged site on the resin. The parameter acts on the step through residence time alone, and prior knowledge places no interaction between residence time and the load chemistry. It is therefore bounded by moving it one factor at a time and is marked “univariate (justified)”. The anion exchange plan PCP-008 records the justification with the design. Table 4.7 gives the failure modes.

Table 4.7: Potential failure modes and their effects, anion exchange.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Load pH	Load pH below its characterized range (below pH 7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal	Host Cell Protein (HCP)
Load conductivity	Load conductivity above its characterized range	Reduced viral clearance (interacts with pH)	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Protein load	Protein load above its characterized range (above 300 g/L resin)	Reduced HCP removal capacity	Host Cell Protein (HCP)
Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)

4.7 Small-Virus Retentive Filtration

The virus filter removes small non-enveloped virus by size and is the principal MVM clearance step. Both of its parameters are scored against MVM clearance, and both reach 700. The step has no parameter that affects process performance alone.

Volumetric load is the parameter on which the step turns. The log reduction is expected to fall as the volume filtered per unit area rises, because the retentive layer accumulates material and the chance of a parvovirus passing rises with it. Filtration pressure is scored against both model viruses, and it sets the driving force across that accumulating layer. Parvovirus filters of this kind can retain less virus at low pressure or low flux and after a pressure interruption, so the direction of a pressure effect on the log reduction is known. Its size for this filter is not. The exposure is largest at high volumetric load, where the layer is thickest. Both parameters are therefore carried into the designed experiment. Table 4.8 gives the failure modes.

Table 4.8: Potential failure modes and their effects, virus filtration.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Filtration volume (load)	Filtration volume (load) above 140 L/m2	Reduced MVM log-reduction (parvovirus breakthrough risk)	Viral clearance — MVM (parvovirus)
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)

4.8 Ultrafiltration / Diafiltration

The formulation step concentrates the antibody and exchanges it into the drug substance buffer. It forms and clears no attribute in Table 3.1. Its 3 parameters are scored against process performance alone and sit at the floor of the ranking at 96. The A-Mab case study gives no characterization of this step (CMC Biotech Working Group 2009), so the assessment rests on platform experience alone.

A number of diavolumes below its range leaves the buffer exchange incomplete. Each diavolume removes a fixed fraction of the buffer species left in the retentate, so the residual falls with the number of diavolumes and not with the time spent filtering. Flux is expected to rise with transmembrane pressure until the membrane reaches the region in which flux no longer depends on pressure. Beyond that point the concentration polarization layer at the membrane surface thickens and flux stops rising, so a transmembrane pressure outside its range is expected to change the processing time. The final concentration is the drug substance concentration itself, which is measured and released against a specification. A failure of this parameter is therefore seen before the batch is released. Each parameter is studied one factor at a time. Table 4.9 gives the failure modes.

Table 4.9: Potential failure modes and their effects, UF/DF.

Parameter	Potential failure mode	Potential effect	Attribute(s) at risk
Number of diavolumes	Number of diavolumes below 7 DV	Incomplete buffer exchange into the formulation buffer	process performance
Transmembrane pressure	Transmembrane pressure outside 10–15 psi	Affects flux/processing (formulation)	process performance
Final DS concentration	Final DS concentration outside 65–85 g/L	DS concentration spec (formulation)	process performance

5 Characterization study assignment

The study assignment is the main output of this assessment. Of the 37 parameters, 22 are assigned to a multivariate design and 15 to univariate study. A designed experiment is assigned at 6 of the 8 unit operations. The remaining 2, harvest and UF/DF, have no parameter scored against a quality attribute, and no designed experiment is assigned to them. Table 5.1 gives the split per unit operation, with the plan and the report that cover each.

Table 5.1: Characterization scope per unit operation, and the documents that cover it.

Step	Unit operation	Parameters	Scored against a quality attribute	Multivariate design	Univariate	Plan and report
3	Production Bioreactor	9	5	5	4	PCP-003 / PCR-003
4	Harvest and Clarification	3	0	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	2	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	5	4	1	PCP-008 / PCR-008
9	Small-Virus Retentive Filtration	2	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration	3	0	0	3	PCP-010 / PCR-010

The parameters assigned to a multivariate design are those judged able to act together with another parameter of the same unit operation. The judgement was made from platform and prior product experience, and Section 4 records it where it decides the study type. The capture step carries 2 parameters into its design although they are scored against process performance alone, so that the same design bounds step yield.

The assessment marks 1 parameter “univariate (justified)”, the substitution that the A-Mab risk ranking and filtering approach permits where the study can be justified (CMC Biotech Working Group 2009). It is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and acts on the step through residence time alone. A parameter that does not act together with the load chemistry can be bounded by moving it one factor at a time, and the anion exchange plan PCP-008 carries that justification with its design.

No parameter is excluded from study on the ground that its risk is low. The lowest ranked parameters change step yield and process consistency, not a quality attribute. Apart from the 2 Protein A parameters carried into the capture design, they are studied one factor at a time, so that the ranges entered in the batch record rest on data and not on platform precedent alone.

Table 5.2 gives the parameter set handed to the plans, with the set-point, the range to be studied and the normal operating range (NOR). Against these NORs the range to be studied is wider at both ends for 31 of the 37 parameters, so that the edges of the characterized region lie outside the region in which the process is run (CMC Biotech Working Group 2009). For 5 parameters the range studied and the NOR share their low end. These are post-clarification turbidity, hold time, filtration volume (load), number of diavolumes and transmembrane pressure. For A-Mab concentration the NOR equals the range studied.

The set-points and NORs in Table 5.2 are those this assessment was made against, and PCMP-001 Version 2.0 Table 4.2 gives the values in force.

Table 5.2: The parameter set handed to the characterization plans, with the set-point, the range to be studied and the NOR.

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
Production bioreactor	Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
	Culture temperature	°C	35	34–36	34.5–35.5	multivariate
	CO2 partial pressure (pCO2)	mmHg	100	40–160	70–130	multivariate
	Osmolality	mOsm/kg	400	360–440	375–425	multivariate
	Culture duration	days	17	15–19	16–18	multivariate
	Dissolved oxygen	% air saturation	50	30–70	40–60	univariate
	Initial viable cell conc.	10^6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
	Basal medium concentration	× standard	1.2	0.8–1.6	1–1.4	univariate
	Nutrient feed-1 volume	% of working volume	12	9–15	10–14	univariate
Harvest	Centrifugation (rcf)	g	9000	6000–12000	8000–10000	univariate
	Depth filter load	L/m2	120	80–160	100–140	univariate
	Post-clarification turbidity[a)]	NTU	5	1–20	1–10	univariate
Protein A	Protein load	g/L resin	30	10–50	20–40	multivariate
	Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
	Load flow rate	cm/hr	200	100–300	150–250	multivariate
	End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
	Operating temperature	°C	22	15–30	18–26	univariate
	Bed height	cm	20	10–30	15–25	univariate
Low-pH viral inactivation	Inactivation pH	pH	3.5	3.2–4	3.4–3.6	multivariate
	Hold time	min	90	60–180	60–120	multivariate
	Temperature	°C	21	15–25	19–23	multivariate
	A-Mab concentration	g/L	20	5–35	5–35	univariate
Cation exchange	Protein load	g/L resin	25	10–40	18–32	multivariate
	Load/Wash conductivity	mS/cm	5	3–7	4–6	multivariate
	Elution buffer pH	pH	6	5.8–6.2	5.85–6.15	multivariate
	Elution stop collect	AU (OD280, descending)	1	0.5–1.5	0.7–1.3	multivariate
	Elution flow rate	cm/hr	200	100–300	150–250	univariate
Anion exchange	Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
	Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
	Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
	Protein load	g/L resin	200	50–300	120–280	multivariate
	Operating flow rate	cm/hr	300	150–450	200–400	univariate[b)]
Virus filtration	Filtration volume (load)	L/m2	90	50–140	50–105	multivariate
	Filtration pressure	psi	13	8–30	10–20	multivariate
UF/DF	Number of diavolumes	DV	8	7–10	7–9	univariate
	Transmembrane pressure	psi	12	10–15	10–14	univariate
	Final DS concentration	g/L	75	65–85	70–80	univariate

[a)] A measured value, not a set one. The set-point and NOR apply to the turbidity measured after clarification.
[b)] Marked univariate (justified) in the risk ranking.
The set-points and NORs are those this assessment was made against. PCMP-001 Version 2.0 Table 4.2 gives the values in force.

The assignment is prospective. It is revisited when the characterization reports are available. A parameter whose measured effect contradicts the judgement made here is reported as such in the report of its unit operation.

6 Assumptions and limitations

The assessment assumes that platform behaviour transfers to A-Mab. The mechanisms it relies on were established on the earlier humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab and on A-Mab clinical and engineering experience (CMC Biotech Working Group 2009). The assumption is that A-Mab responds to the same parameters in the same directions. Where it does not, the characterization study finds it, which is one reason the ranges studied are wider than the NORs.

The parameter list is assumed complete. It is the list in the process description transferred under PTP-001, and a parameter that is not in that description is not assessed. An equipment or facility difference at the receiving site that introduces a new parameter is handled under the transfer plan and requires this assessment to be revised.

Severity is taken from the quality attribute assessment and is not re-derived. That assessment scores impact and uncertainty and deliberately excludes manufacturing capability and detectability (CMC Biotech Working Group 2009), so a severity in Table 3.1 says nothing about how likely the process is to produce a failure.

Occurrence and detection are assigned by class, so the initial RPN takes only 5 values across 37 parameters. Ranking within a unit operation should therefore be read from severity, from the class of the attribute at risk and from the reasoning in Section 4.

The judgement on which parameters act together is the weakest part of the assessment. A designed experiment estimates interactions only among the parameters it carries. An interaction between a parameter in the design and one left out of it cannot be found by that design. The clearest exposure is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and studied one factor at a time. No study as assigned can detect an interaction of residence time with load pH and conductivity. The univariate result for flow rate therefore bounds the step only if that interaction is absent, as prior knowledge expects, and the anion exchange plan PCP-008 carries that justification.

Interactions between unit operations are not scored. Each parameter is assessed against the attributes it can affect at its own step. Material does carry properties from one step to the next. The clearest case is the elution pH of the capture step, which sets the starting pH of the low-pH hold. That link is handled in the receiving step, where the pool is titrated to the set-point of inactivation pH, and not by a score in Table 4.1. An attribute carried from one step to the next is judged against the intermediate acceptance criteria of PCMP-001 Version 2.0 §7.3, which carry each drug substance criterion back through the steps downstream.

Fragments are outside this assessment, for the reasons given in Section 3.

Finally, no number in this document is a measurement of A-Mab process behaviour. The scores are judgements, the studies had not been run when the assessment was made, and the ranges in Table 5.2 are ranges to be studied and not ranges shown to be acceptable.

7 Outputs and downstream use

This assessment produces three outputs. The ranking in Table 4.1 is the pre-characterization risk basis. The study types in Table 4.1 and the split in Table 5.1 are the scope. The parameter set in Table 5.2 is what each plan takes up.

Per unit operation, the consumer is the characterization plan named in Table 5.1. Each plan cites this assessment as its risk basis and takes the study type and the range to be studied. It adds the scale-down model, the analytical methods, the sampling plan and the acceptance criteria. The scale-down model and the acceptance criteria follow PCMP-001 Version 2.0 (§7.1 and §7.3). The report of the unit operation measures the effects, defines the operating region and classifies each parameter under SOP-4001.

PCMP-001 consumes the assessment at campaign level. It states once what the plans would otherwise each restate, including the number of parameters assigned to multivariate and to univariate study across the train (PCMP-001 Table 4.1). PCMR-001 consolidates the reports and is where the risk that remains after characterization is addressed.

This assessment is changed only by revision. A change to the parameter list, to the severity of an attribute or to an assigned study type is made by revising this document, and the affected plans are revised with it. Version 2.0 changes no score and no study type (Section 1.1).

8 Approvals

This assessment is approved by the functions in the approval record at the front of the document, with the responsibilities given in Table 8.1. Approval of Version 2.0 confirms the changes listed in Section 1.1 and confirms that the method, the scores and the study assignment are those of Version 1.0.

Table 8.1: Responsibilities of the approving functions.

Function	Responsibility for this assessment
Process Development / MSAT	Author. Owns the parameter list, the failure modes, the scores and the study assignment
Analytical Development	The attribute register and the methods that measure each attribute, including the fragment method of AMD-001
Manufacturing Science	Completeness of the parameter list against the process description transferred under PTP-001
Statistics	The scoring scales and the assignment of parameters to designed experiments
Quality Assurance	Approval, and closure of INV-001 corrective action CA3, which reissues this document at Version 2.0

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